

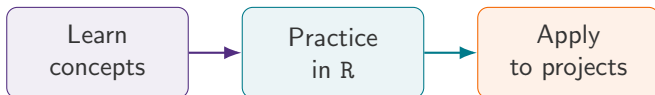
DSA 8070: Intro to Applied Multivariate Analysis

Course Information

Whitney Huang

Clemson University

How this course is organized



Key idea

The course emphasizes interpretation, computation, and responsible application of multivariate methods

About the Instructor

- ▶ Associate Professor of Applied Statistics and Data Science
- ▶ Born in Laramie, Wyoming, and grew up in Taiwan
- ▶ B.S. in Mechanical Engineering; transitioned to Statistics in graduate school
- ▶ Ph.D. in Statistics, Purdue University, 2017



Why I mention this

Multivariate analysis is used across engineering, science, business, and data science; the course is designed with that interdisciplinary perspective in mind

How to Reach Me

- ▶ **Email:** wkhuang@clemson.edu
- ▶ **Office:** O-221 Martin Hall
- ▶ **Weekly office hour:** Wednesday, 8:00–9:00 PM ET, via Zoom

Communication

Email is the best channel for individual questions. **Please include DSA 8070 in the subject line of your email.** Use Canvas announcements and modules for course-wide information and materials

Course Structure

Weekly learning	Major applications
▶ Lecture slides / videos ▶ R-based practice ▶ Weekly R labs	▶ Project I ▶ Project II ▶ Project III

Key idea

The projects are designed to integrate statistical reasoning, computation, interpretation, and communication.

Important Logistics

- ▶ There will be **three projects**:
 - Project I: Sep. 24
 - Project II: Nov. 5
 - Project III: Dec. 3
- ▶ Weekly R Labs are submitted through Canvas by 11:59 PM ET on the due date.
- ▶ The lowest R Lab grade will be dropped.
- ▶ No lectures during Thanksgiving week (Nov. 23–27).

Always use Canvas for the most current deadlines and announcements.

Course Materials on Canvas

Course information

- ▶ Syllabus
- ▶ Announcements
- ▶ Weekly modules

Learning materials

- ▶ Slides and videos
- ▶ R Labs and projects
- ▶ Data sets and supporting files

whitneyhuang83.github.io

Reference Books

- ▶ Daniel Zelterman (2015), *Applied Multivariate Statistics with R*.
- ▶ Alan Izenman (2008), *Modern Multivariate Statistical Techniques: Regression, Classification, and Manifold Learning*.
- ▶ Alvin Rencher and William Christensen (2012), *Methods of Multivariate Analysis*, 3rd ed.
- ▶ Richard Johnson and Dean Wichern (2008), *Applied Multivariate Statistical Analysis*, 6th ed.

How to use the books

No single text is followed page-by-page. Readings are used as references that complement the lecture notes and R examples.

Evaluation

Weights	
Industrial Spotlight	5%
R Labs	20%
Project I	25%
Project II	25%
Project III	25%

Letter grades	
90.00–100	A
88.00–89.99	A-
85.00–87.99	B+
80.00–84.99	B
78.00–79.99	B-
75.00–77.99	C+
70.00–74.99	C
68.00–69.99	C-
≤ 67.99	F

Computing: R and RStudio

We will use **R** and **RStudio** throughout the course.

- ▶ Free and open-source tools for statistical computing
- ▶ Used to reproduce lecture examples
- ▶ Used in R Labs and project analyses



www.r-project.org

RStudio

Quick check

Before Week 1, can you open RStudio and run a simple command such as `mean(c(1,2,3))`?

Topics and Course Progression

Week	Topic
1	Introduction and multivariate data exploration
2	A short review of matrix algebra
3	Multivariate normal distributions and copula models
4	Inference and comparison of mean vectors
5–6	Multivariate regression
7	Inference for covariance matrices
8	Principal component analysis
9	Factor analysis
10	Canonical correlation analysis
11	Discrimination and classification
12	Cluster analysis
13	Multidimensional scaling and distance embedding
14	Manifold learning and nonlinear embedding
15	Thanksgiving week – no class
16	Review

Key idea

The course moves from data representation and probability models to inference, prediction, dimension reduction, and unsupervised learning.